

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE:MLA02

COURSE NAME: Fundamentals Of Machine Learning

COURSE OUTCOME COVERED

CO1: Analyse machine learning fundamentals and mathematical foundations, including probability theory, Bayes decision theory, and information theory. (BL3)

Assessment Tool Weightage:20%

QUESTIONS: 10

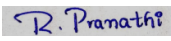
Total Marks:50

Annexure A

Concept Mapping

Code of Conduct:

*I **R.Pranathi(192525076)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: 

Annexure A

Concept Mapping

Q1 Construct a concept map integrating machine learning foundations with real-world applications such as spam detection or recommendation systems. (5Mark- 7 Min)

Q2 Develop a concept map illustrating how random variables and probability distributions are utilized in predicting uncertain outcomes. (5Mark- 7 Min)

Q3 Formulate a concept map demonstrating the application of Bayes decision theory to medical diagnosis and decision-making problems.(5Mark- 7 Min)

Q4 Design a concept map applying the stages of a learning system to address a student performance prediction problem. (5Mark- 7 Min)

Q5 Synthesize a concept map representing how information theory concepts are employed to quantify uncertainty in machine learning datasets. (5Mark- 7 Min)

Q6 Construct a concept map linking supervised, unsupervised, and reinforcement learning with their practical applications in intelligent systems. (5Mark- 7 Min)

Q7 Develop a concept map illustrating the relationship between random variables, conditional probability, and Bayes theorem in spam classification. (5Mark- 7 Min)

Q8 Formulate a concept map demonstrating how probability distributions support decision-making in weather forecasting systems. (5Mark- 7 Min)

Q9 Design a concept map applying the workflow of a machine learning system to fraud detection in banking transactions. (5Mark- 7 Min)

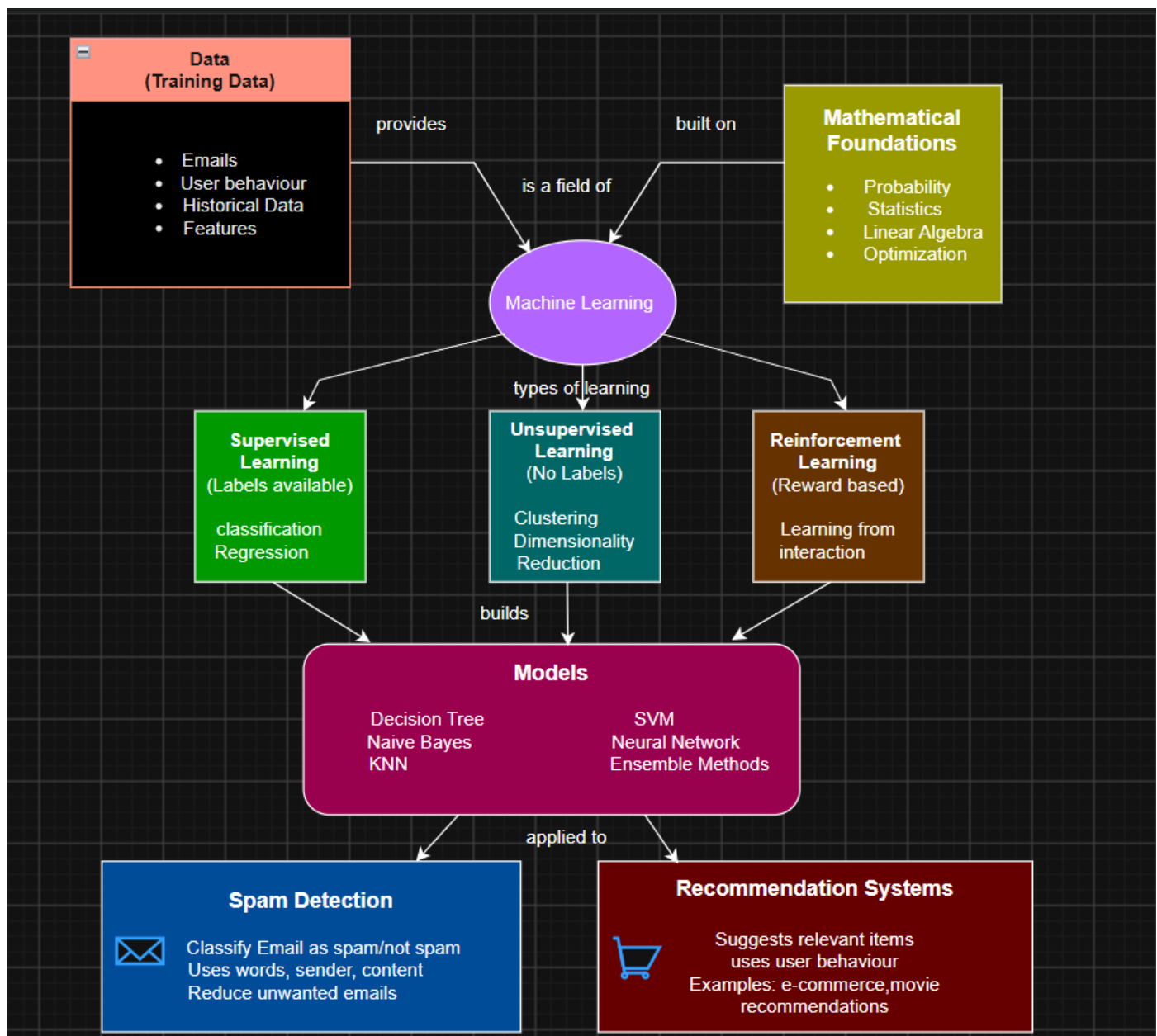
Q10 Synthesize a concept map representing the role of entropy, information gain, and uncertainty reduction in classification problems. (5Mark- 7 Min)

RUBRICS FOR CALCULATION

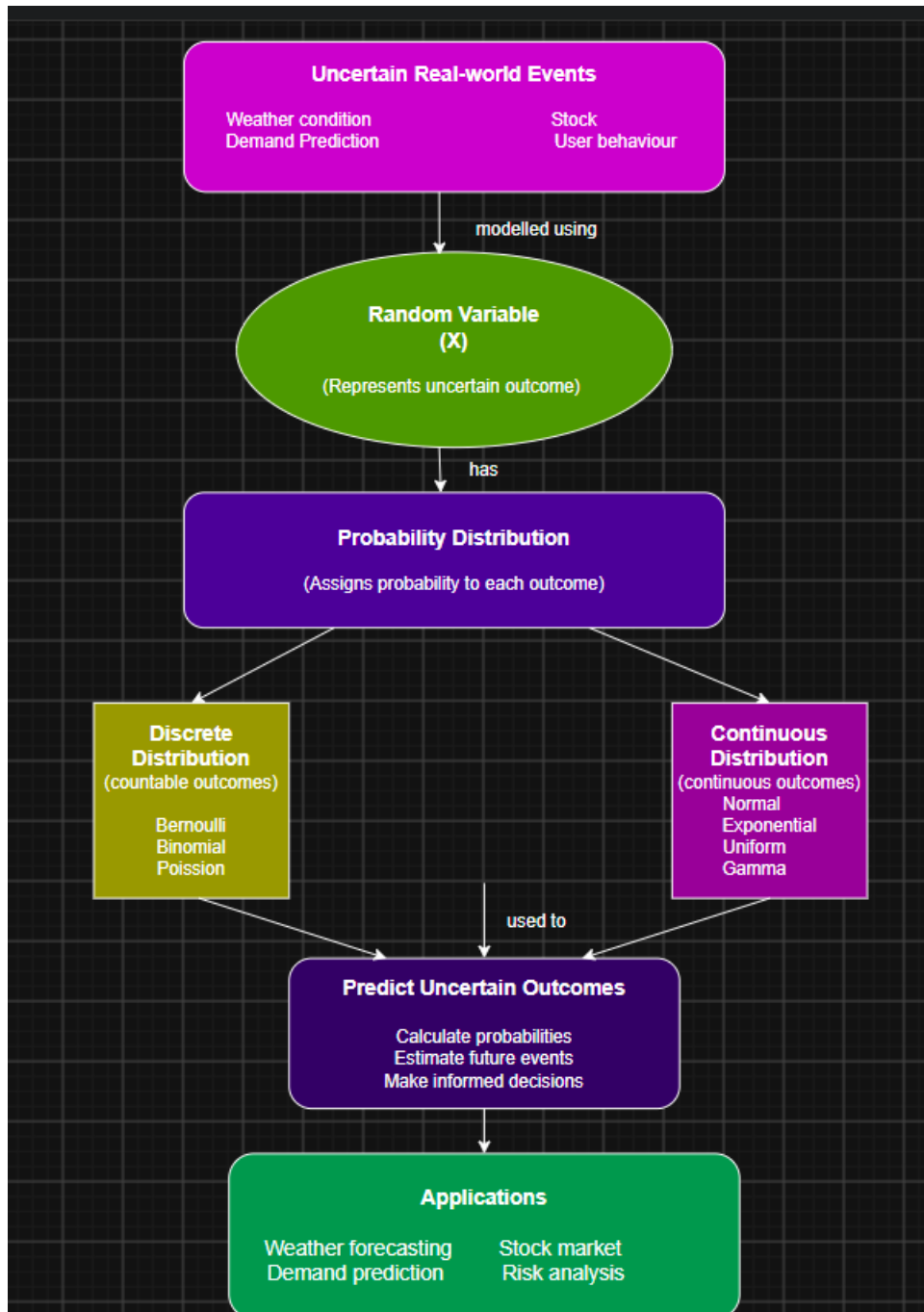
Concept Mapping					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure

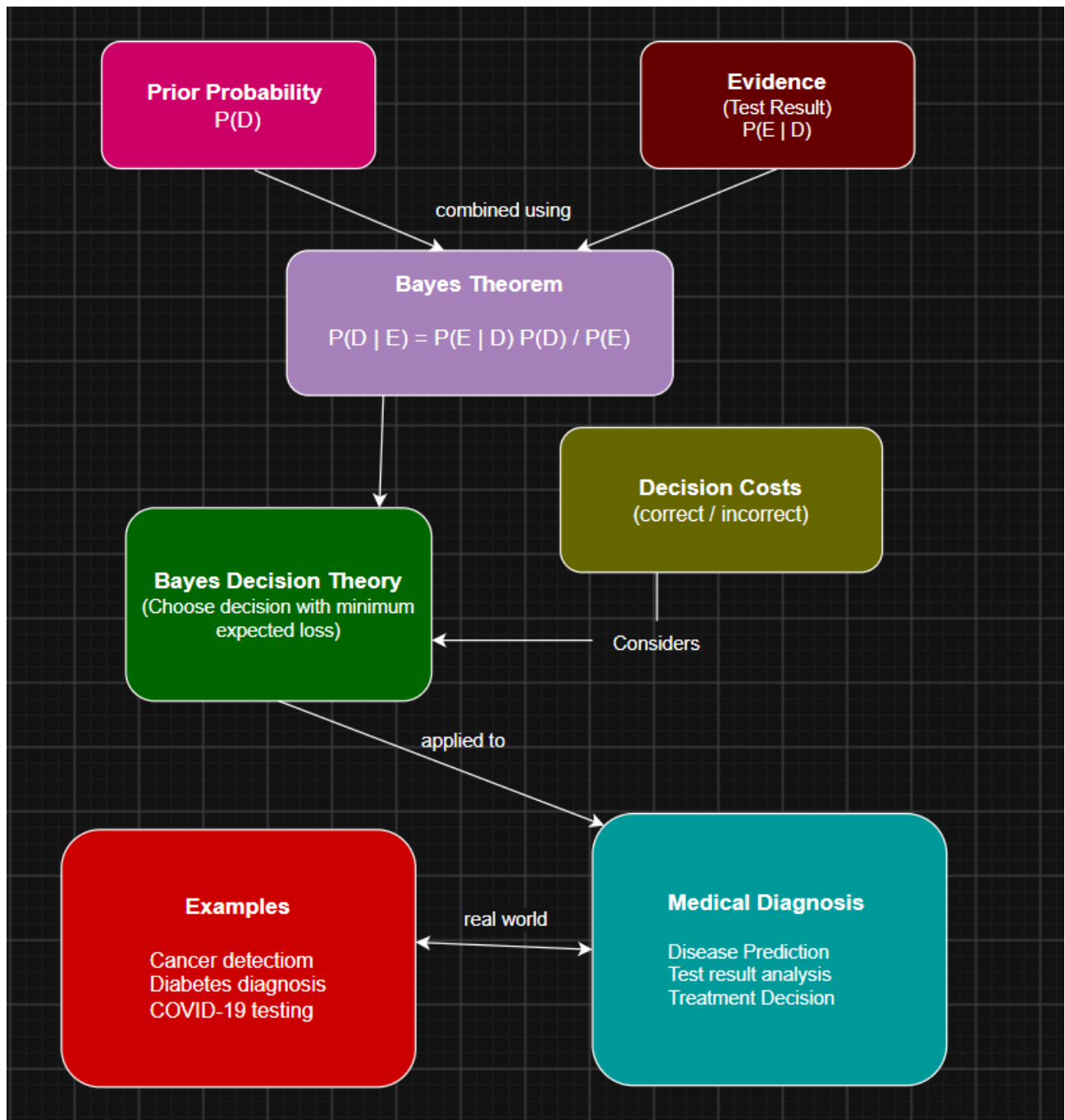
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

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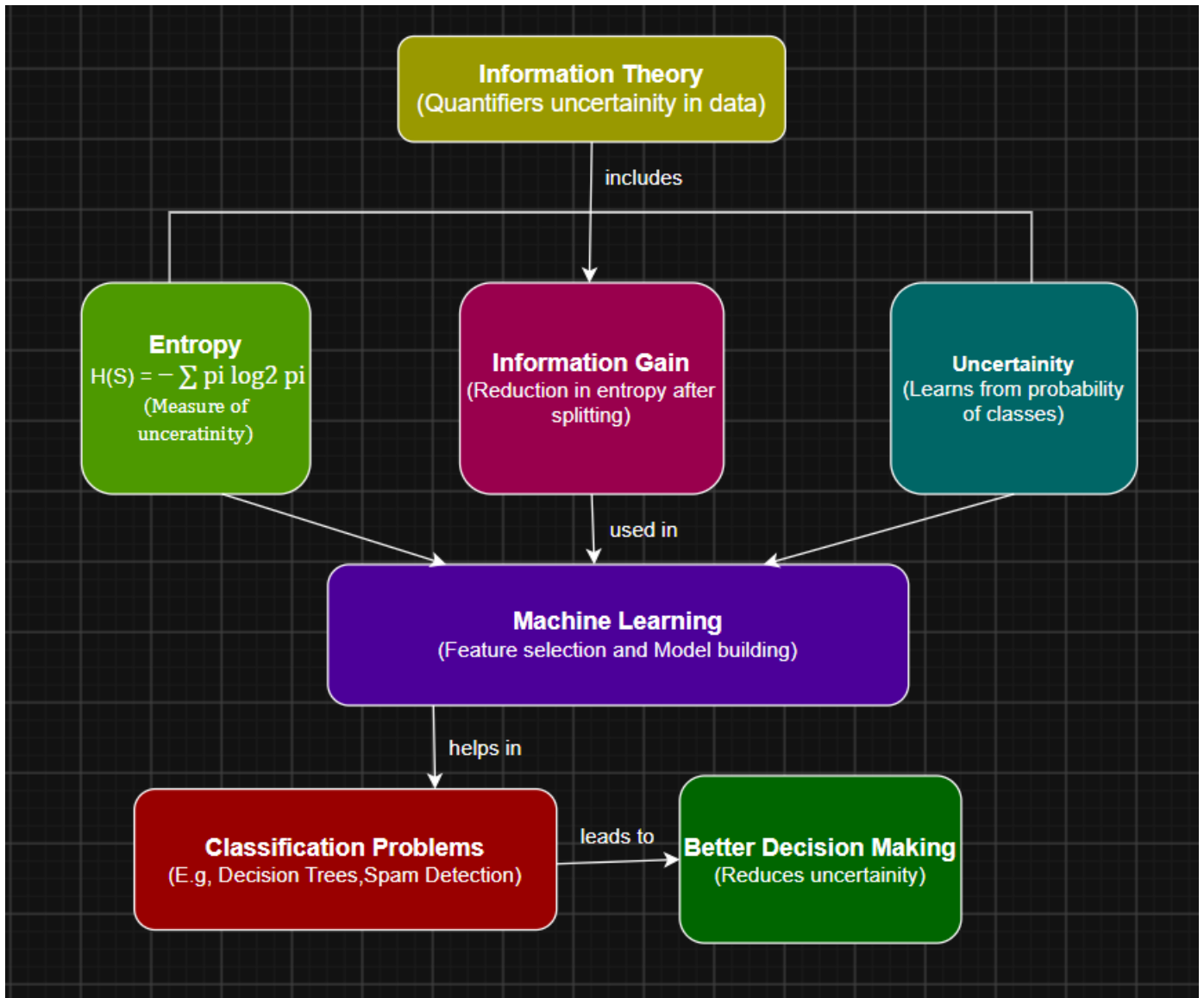




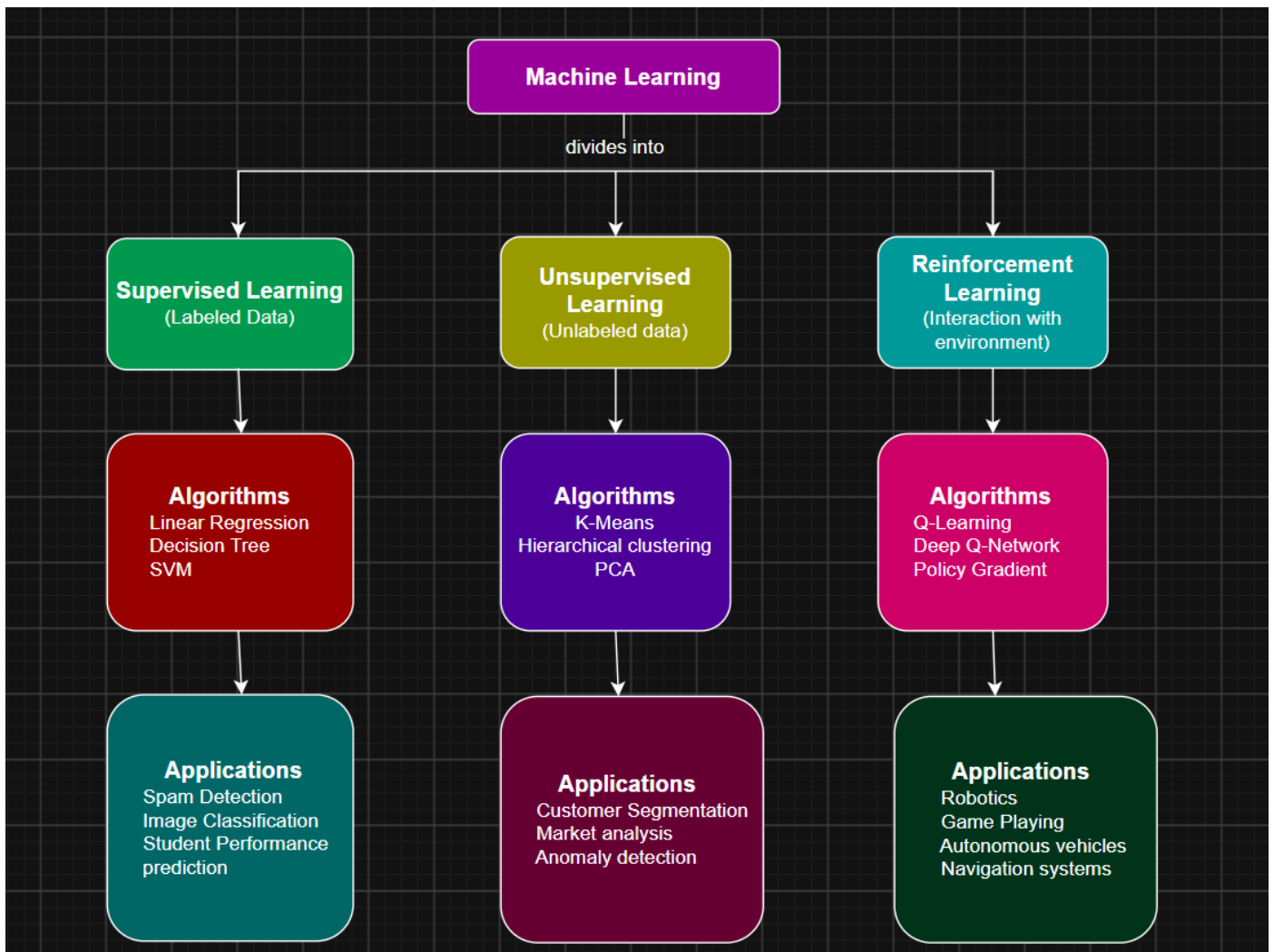
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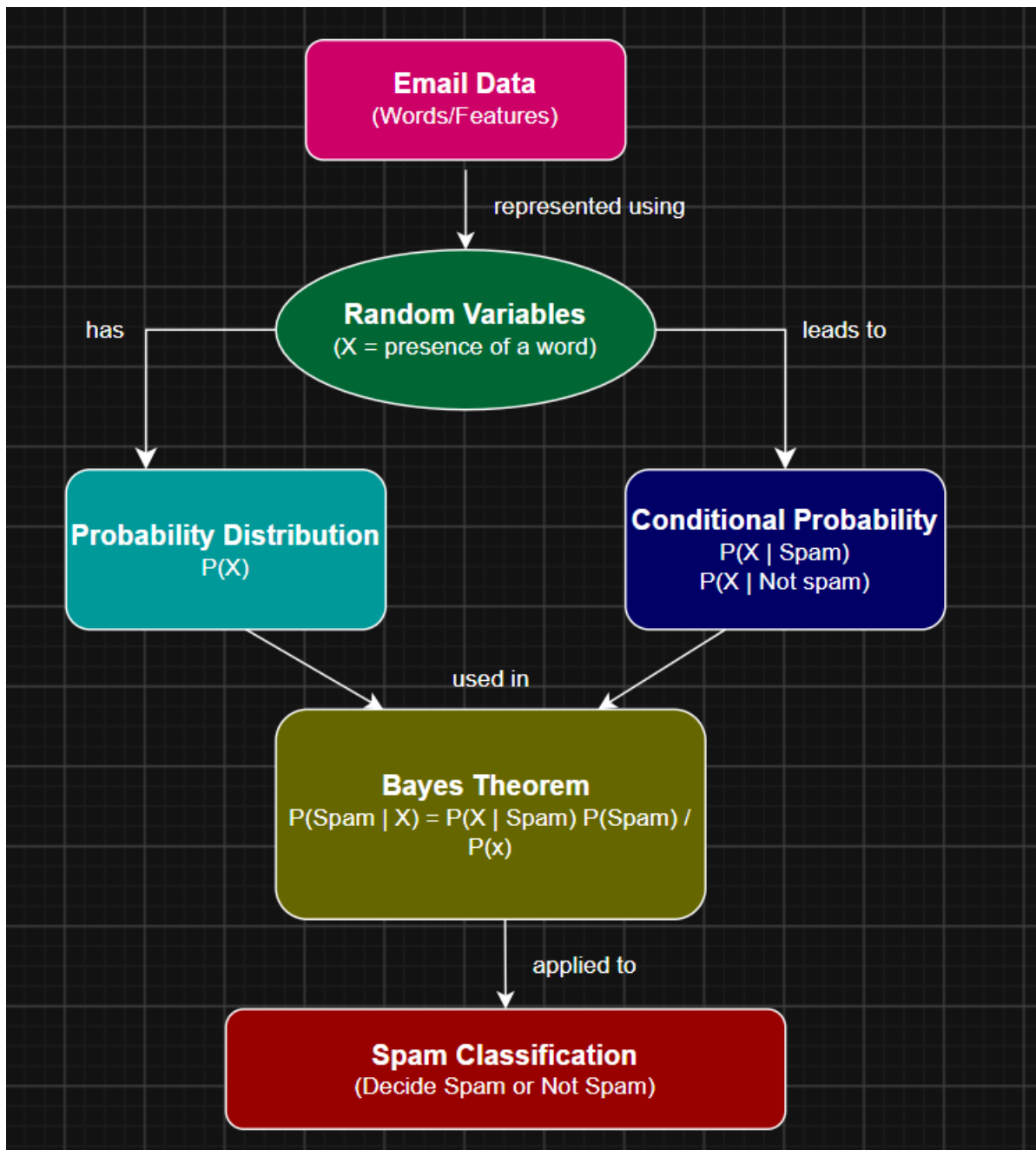
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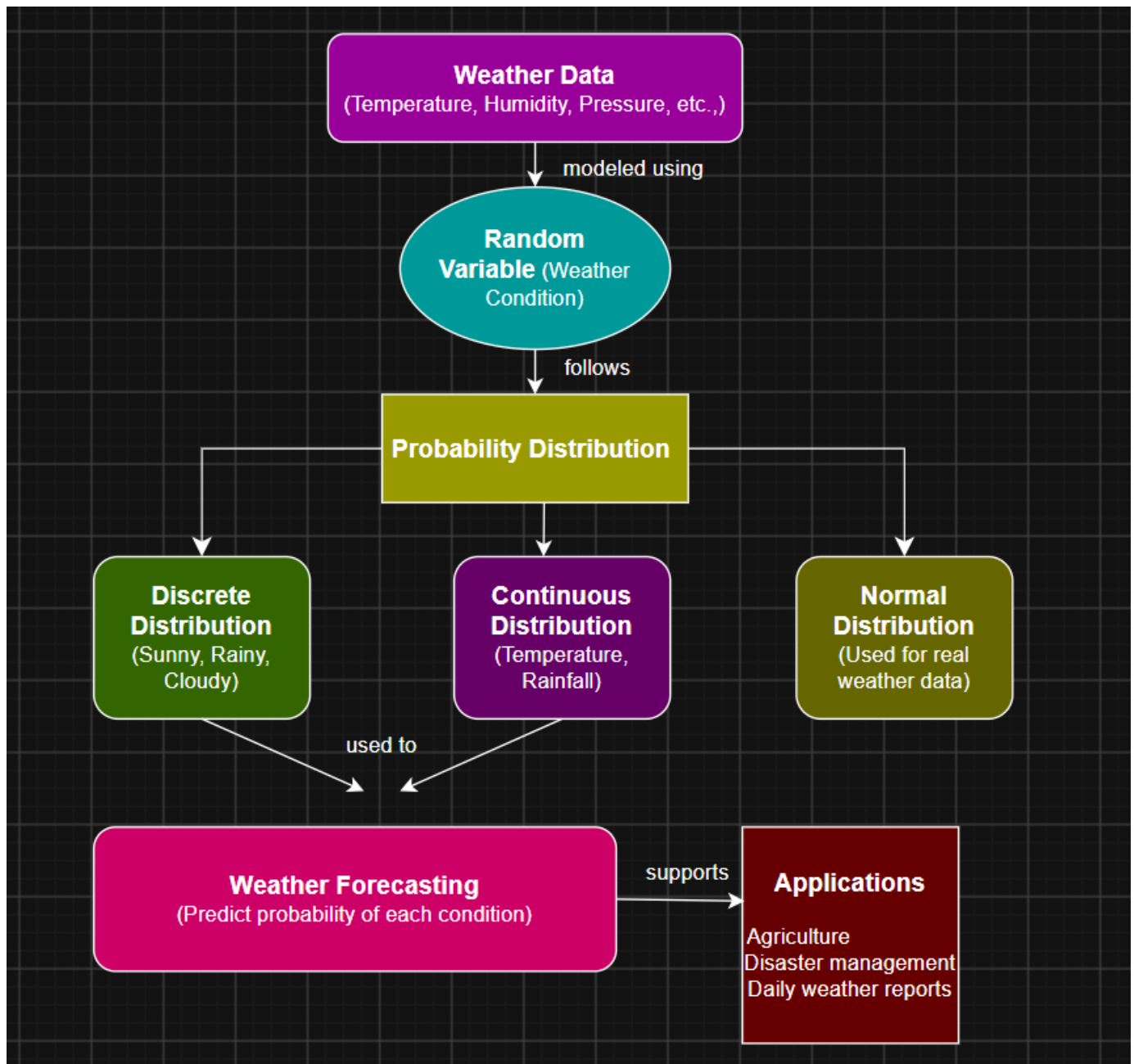
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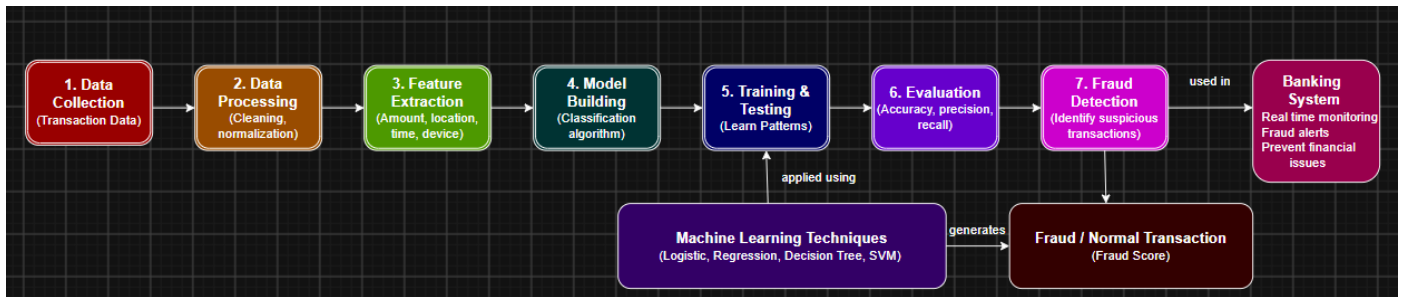
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